AAD-CEMS-200 NDIR ONLINE CEMS GAS ANALYZER

USER MANUAL

(NDIR-CEMS/2023/001-01)

Gas	Method	Range
Carbon Monoxide	NDIR	0-5000 ppm
Sulphur Dioxide	NDIR	0-5000 ppm
Nitric Oxide	NDIR	0-5000 ppm
Oxygen	Zirconia	0-25%
Carbon Monoxide	NDIR	0-25%
Hydrogen Chloride	Electrochemical	0-100 ppm

CONTENTS

1. INTRODUCTION

Continuous Emission Monitoring Systems are used to monitor flue gas for the presence of Carbon Monoxide (CO), Sulphur Di-Oxide (SO2), Oxides of Nitrogen (NOx), Carbon Di-Oxide (CO2), Oxygen (O2) and Suspended Particulate Matter (SPM). CEMS analyzers provide crucial information on the industrial combustions and are used to tune and control emissions.

Flue gases are one of the chief causing factors of Acid Rain and their composition of hazardous gases shall be controlled as per CPCB guidelines. In addition to afore-mentioned gases, other gases such as Ammonia (NH3), Hydrogen Chloride (HCl), Hydrogen Flouride (HF), Methane (CH4), etc are also measured using CEMS analyzers.

Apart from Oxygen, all the other gases are measured using Non-Dispersive Infra-Red (NDIR), Laser Absorption Spectroscopy (LAS), Tunable LAS (TDLAS), Electro Chemical (EC), Chemiluminescence (CLD), and Photo Acoustic (PAS) sensors. Oxygen is measured using EC, Paramagnetic or Zirconia measurement. In these gases, CO2 and O2 are measured in percentage (%) while the others are measured in parts per million (ppm) or parts per billion (ppb). SPM is estimated by measuring the opacity of the flue gas.

In India, CEMS are used as a means to comply with air emission standard defined by the Central Pollution Control Board (CPCB) and State Pollution Control Boards SPCB’s). Facilities employ the use of CEMS to continuously collect, record and report the required emissions data.

2. GENERAL SAFETY & PRECAUTIONS

Avoid unsafe acts and conditions. Do not operate or install the instrument before reading this chapter. Installing the AAD-CEMS-200 involves wiring high voltage equipment. Installation must be performed by someone who understands the dangers of, and is qualified to, wire electrical devices. If information or instructions are not clear, DO NOT PROCEED until clarification can be obtained. Follow all safety instructions as given below.

2.1. Follow basic safety rules when working with or near high voltage circuits. The AAD-CEMS-200 is a high voltage instrument using 220V/110V. Peripheral devices are also high voltage devices. Keep away from live circuits.

2.2. Do not change parts or make adjustments inside the equipment with high voltages on.

2.3. Do not service alone.

2.4. Do not energize equipment if there is any evidence of water leakage.

2.5. All wiring that involves connections to mains power must be performed by a qualified licensed electrician, and must conform to all locally applicable electrical codes.

2.6. Do not make connections while power is applied. Turn off power and assure power "Lockout" before installing or servicing to avoid contact with electrically powered circuits. This includes working on devices connected to the relay outputs and auxiliary input systems.

2.7. Disconnect external power to AAD-CEMS-200 before connecting or disconnecting components and/or peripheral devices.

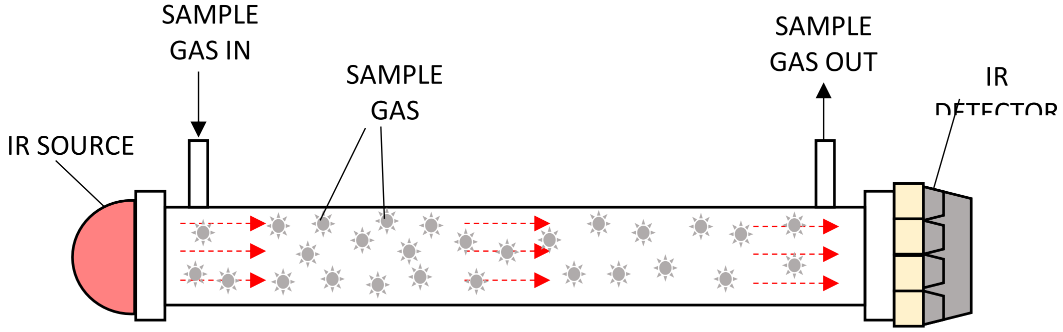
2.8. Install and use the AAD-CEMS-200 only in accordance with information documented in this manual.

3. AAD-CEMS-200 OPERATING PRINCIPLE

AAD-CEMS-200 CEMS analyzers use Hot Extraction method to extract the flue gases for measurement. A sampling probe with built-in mesh filter is installed in the stack to extract the flue gas. The sample gas is carried along a heated sample line at high temperature into a sample conditioning unit. The sample is filtered to remove particulate matter and dried, usually with a chiller, to remove moisture. The extractive type CEMS systems cool the sample under controlled conditions to condense the water out and dry the sample. Once the sample has been cooled and dried, it is then sent to an analyzer. This method is referenced as the cool, dry or just dry basis analysis. But some of the gases are either partially or fully water soluble. By cooling the sample and removing the water for a cool dry analysis, some of these gasses that have dissolved in the moisture can also be removed from the gas as the water is condensed. An error can be induced on gasses such as SO2 and NO2. Along these same lines, it is impossible to measure highly water-soluble gasses like NH3, HCl and HF.

3.1. THEORY OF OPERATION

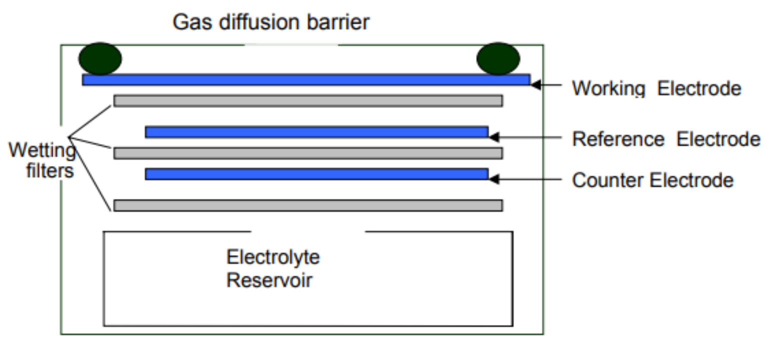
3.1.1. NDIR Sensors



AAD-CEMS-200 operates under NDIR spectroscopy technique. Over 100 different gases can be detected by NDIR sensors from ppm to percent range. In many fields of applications, the sensors are classified as standard method because the measurement method is non-contact and wear-free. In NDIR spectroscopy the sample gas under consideration is filled in a cuvette of a defined length and is irradiated by infrared light. The length of the cuvette is chosen in such a way that the range requirements can be fulfilled. The gas molecules present in the sample gas absorb the infrared light at specified wavelengths (absorption spectra) according to their type. The absorption spectra is analysed to identify the presence of gas molecules and their concentration.

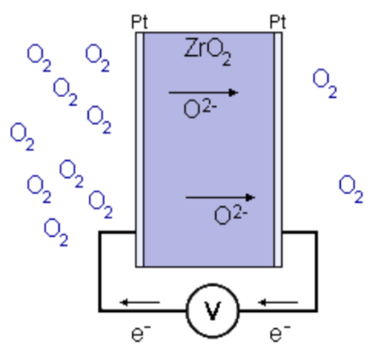
The infrared light is generated by a semiconductor based Broadband IR source and is attached to one end of the cuvette. The other end of the cuvette is attached with a semiconductor based multi band IR detector. Bandpass filters which suit for the gases are used to restrict the sensor from responding to every wavelength. In the ideal case, the target gases should alone absorb the light of their wavelengths and no other gases in the sample gas mixture absorb the light. The resulting intensity of infrared light at the other end of the cuvette should be measured to calculate the absorption spectra. Absorption ranges can also overlap which results a cross sensitivity. In order to avoid distortion in the measurement results cross sensitivity has to be compensated or avoided by a skilful choice of the frequency bands.

3.1.2. Electrochemical Sensors



Electrochemical sensors detect gas concentration by measuring current based on the electrochemical principle, which utilizes the electrochemical oxidation process of target gas on the working electrode inside the electrolytic cell, the current produced in electrochemical reaction of the target gas are in direct proportion with its concentration while following Faraday law, then concentration of the gas could be get by measuring value of current.

3.1.3. Zirconia Sensors



At the core of the oxygen sensor is the sensing cell. The cell consists of two zirconium dioxide (ZrO2) squares coated with a thin porous layer of platinum which serve as electrodes. The platinum electrodes provide the necessary catalytic effect for the oxygen to dissociate, allowing the oxygen ions to be transported in and out of the ZrO2. Two different ion concentrations on either side of an electrolyte generate an electrical potential which is proportional to concentration of Oxygen.

3.1.4. SPM through Opacity Measurement

AAD-CEMS-200 uses an in-situ, non-contact, cross-duct, optical laser dust measurement technique which measures dust concentration by measuring opacity of the transmission medium. These monitors are designed for measurement across stacks, pipes and ducts with diameters from 0.5 m to 10 m. AAD-CEMS-200 uses a transmitter/receiver configuration to measure the opacity/dust concentration along the line of sight. AAD-CEMS-200 uses the principle of light attenuation due to scattering and/or absorption in which a laser light transmitted by the transmitter is scattered by dust particles. The incident light is then collected by a solid-state optical detector for dust quantification.

Opacity can be explained as follows. If a beam of light with frequency *v,* travels through a medium with opacity k_ν, and mass density ρ, both constant, then the intensity will be reduced with distance *x* according to the formula

$$I(x) = I_0 e^{-k_\nu \rho x}$$

where

x is the distance the light has travelled through the medium

I(x) is the intensity of light remaining at distance *x*

I_0 is the initial intensity of light, at

For a given medium at a given frequency, the opacity has a numerical value that may range between 0 and infinity, with units of length^2/mass.

Opacity in air pollution work refers to the percentage of light blocked instead of the attenuation coefficient (aka extinction coefficient) and varies from 0% light blocked to 100% light blocked.

$$Opacity = 100\% \times \left(1 - \frac{I(x)}{I_0}\right)$$

Since AAD-CEMS-200 utilizes a simpler approach for measuring dust concentration, it has no moving parts reducing maintenance costs. Opacity monitor's optimal use is for stack up to 20 m.

AAD-CEMS-200 dust monitoring feature has the following benefits

- Fast response time
- Reasonable cost
- Process control
- No requirement of sample conditioning

Typical applications of optical dust monitor include power plants, cement factories and tunnels. The alignment of the beam is easy as the beam is narrow.

3.2. AUTO SEQUENTIAL SAMPLER

AAD-CEMS-200 has a built-in Auto Sequential Sampler setup that handles Reverse Purge, Vent and Sample modes of operation. As the sample gas is continuously drawn by the sampling pump, dust particles in the gas gets deposited in the inner surface of the sampling tube and the sampling probe. In order to remove the dust, compressed air shall be sent in the reverse direction to push the dust particles back into the stack periodically. This process is called as Reverse Purging. Immediately after the reverse purging operation, the built-up pressure (if any) in the sample hose shall be removed by the vent process. The auto sequential mechanism is controlled by a Programmer Logical Controller. The default timing of the operation is given in the table below. The duration of each operation is reconfigurable as per stack requirement.

State	Duration	Mode
T1	240 s (4 min)	Reverse Purge
T2	15 s	Vent
T3	2s	Change Over
T4	1500 s (25 min)	Sample
T5	2 s	Change Over

3.3. DEHUMIDIFIER

In a CEMS, the presence of a high moisture level in a sample causes a loss of analytes due to artifact formation or absorption. This issue brings about a bias in the measurement data. Thus, moisture removal is an important pretreatment step. Condensation and permeation methods have been widely employed to remove moisture from the CEMS for gaseous compounds. In AAD-CEMS-200, the sample is dried using the condensation method in which the sample gas is passed through a cooling chamber. The cooling chamber uses a thermoelectric cooler with heat radiator.

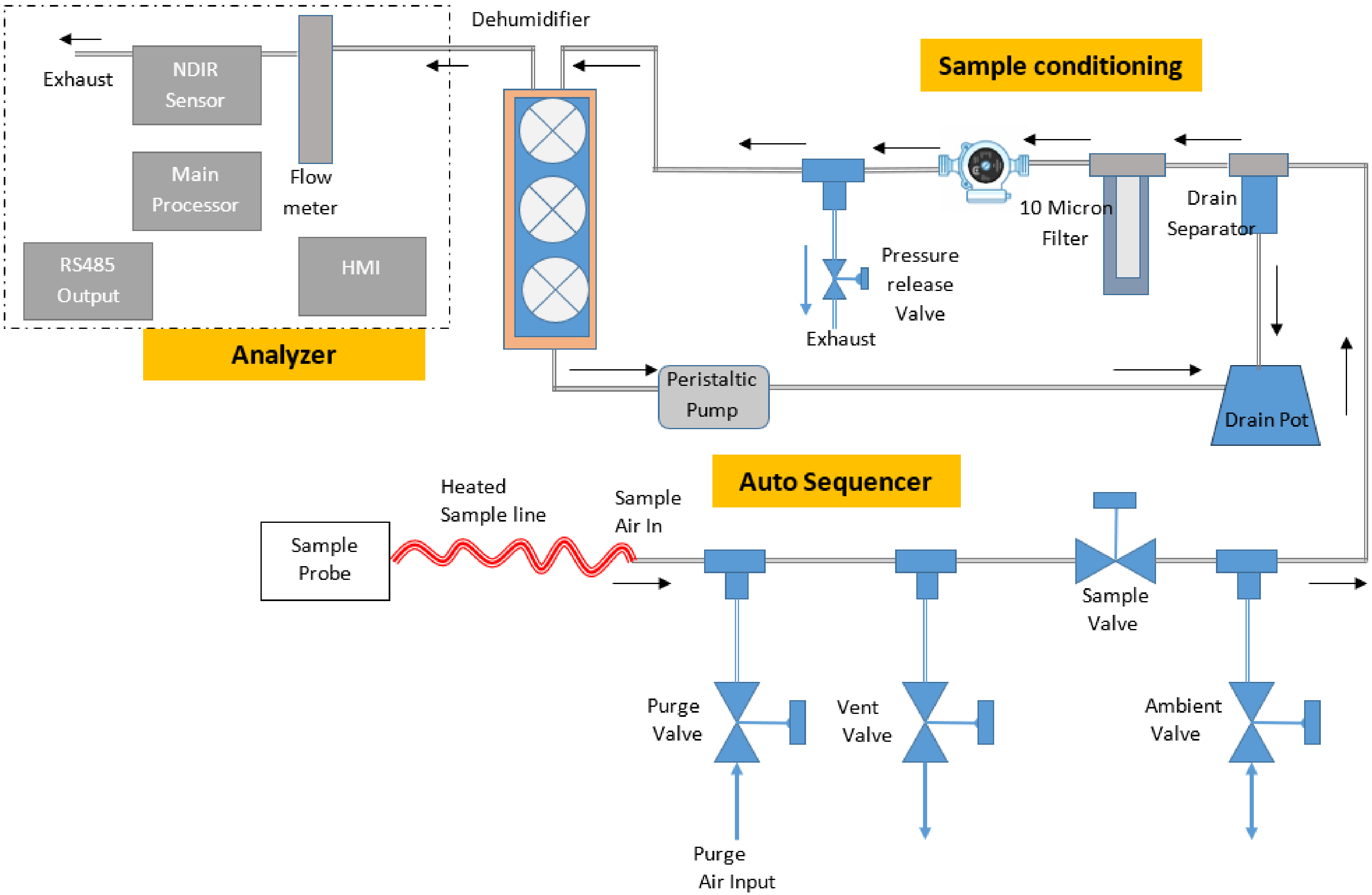
3.4. VACUUM SAMPLING PUMP

An oil free vacuum sampling pump is used to draw the sample gas from the stack. It consists of a pneumatic valve that directs compressed air back and forth between the two sides of the pump.

3.5. FEATURES

- Cost-optimized measurement method due to its modular and compact design
- High quality cell made of special coated cuvette
- Smart temperature management prevents condensation
- No moving parts inside the sensor, long lifetime
- Pressure compensation
- H2O compensation
- Built-in PLC controlled configurable Auto sequential sampler for purging
- High power Thermo-electric cooler for Dehumidification
- Oil free Diaphragm Pump to draw Sample gas
- Built-in controller for heated sample hose
- 7 inch HMI display

4. AAD-CEMS-200 GAS SAMPLING SCHEMATIC DIAGRAM



5. SPECIFICATIONS

Gases and Ranges	Gas	Range	Type
	CO	0-6249 mg/m^3 (0-5000 ppm)	NDIR
	NO	0-6695 mg/m^3 (0-5000 ppm)	NDIR
	SO2	0-14294 mg/m^3 (0-5000 ppm)	NDIR
	CO2	0-25%	NDIR
	HCL	0-100 ppm	EC
	O2	0-25%	Zirconia/EC
SPM	Range: 0-2000 mg/m^3		
Limit of Detection	< 0.5 % (FS)		
Linearity	± 1 % (FS)		
Accuracy	± 1 % (FS)		
Zero Drift	< LOD in 8 hrs		
Span Drift	± 1 % (FS) in 8 hrs		
Rise Time	≤ 10 s		
Warmup Time	≤ 30 minutes		
Operating Temperature	10 – 40° C		
Ambient Humidity	10 – 90 % R.H.		
Gas Pressure	800…1200 mbar (compensated)		
Gas Flow	1000 ml/min		
Digital Resolution	1 mg/m^3 , 1 ppm, 0.1 %		
Sensor Lifetime	More than 5 years for NDIR & Zirconia, 2 years for EC		
Analyzer Specifications			
Display (HMI)	7.0 inch display with resistive touch screen		
Gas Pump	PTFE coated Micro Diaphragm Gas pump (Flow 15 LPM max.)		
Gas Dryer method	Multistage Thermoelectric Cooler with exclusive controller		
Dew point for Cooler	5 ° C		
Drain Separator	Automatic mode – Peristaltic Pump with replaceable hose		
Gas Filter	Hydrophobic, Chemical Resistant, 1 micron, single/multistage filter		
Reverse Purging	Industrial controller based Automatic Sequencer built-in		
Communication Ports	RS485 (Modbus RTU)		
Alarms	Relay NO 24V@5A/240VAC@1A		
Calibration	Manual mode (using HMI)		
Calibration Gas Input	1. Zero gas: N2 gas 2. Span gas: Mixed Gas or Separate gas cylinders		
Power Supply	AC220V ±10% 50 Hz		
Enclosure	Industrial Heavy duty Enclosure with castor wheels		
Dimensions	1900x600x800 (HxWxD) approx		

6. CONNECTORS PIN ASSIGNMENT

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CON3 -Pin No	Configuration
1	NC
6	AI1
2	AI2
7	AI3
3	AI GND
8	RS 485 B-
4	RS485 A+
9	NC
5	NC

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3 pin Connector Pin number	Configuration	Configuration
1	White 24V	Red 24V
2	Brown GND	Black GND
3	Green 5V	Yellow 5V

2 pin Connector Pin number	Configuration	Configuration
1	White RS485 A+ PLC	Red RS485 A+ PLC
2	Brown RS485 B- PLC	Black RS485 B- PLC

7. MODBUS CONNECTION

AAD-CEMS-200 provides MODBUS-RTU communication port through standard RS232 port at a fixed baud rate of 9600 with 8-N-1 bit format. Optional RS485 Level converter is provided for long distance communication. The dust concentration values are available in the Holding registers starting from address 40001 and can be read using Function Code 03 with standard 16-bit-CRC. The MODBUS registers and their addresses are as follows.

S.No	Register Address	Measured Parameter	Unit	Format
1.	40001	CO	mg/m^3	Swapped Float
2.	40003	SO2	mg/m^3	Swapped Float
3.	40005	NOx	mg/m^3	Swapped Float
4.	40007	O2 (Zirconia)	%	Swapped Float
5	40009	CO2	%	Swapped Float
6	40011	O2 (EC)	%	Swapped Float
7.	40013	SPM	mg/m^3	Swapped Float
8.	40015	HCL	ppm	Swapped Float

8. INSTALLATION REQUIREMENTS

The following utilities are required for the installation of EMS2400 dust monitor.

8.1. OPERATING CONDITIONS

The installation site should be free from vibrations, dust, heat and electromagnetic generating sources. The instrument display unit should not be exposed to the direct sunlight.

8.2. POWER SUPPLY

The Instrument operates on 220/110 V ± 10%, 50/60 Hz power supply. UPS power supply is recommended. Two numbers of 5 Amps power socket should be available near to the instrument. Always use properly grounded power supply. The sensitivity and stability of the AAD-CEMS-200 will be impaired if it is not grounded. To avoid possible electrical shock or damage to the equipment, connect earth ground to the AAD-CEMS-200 power supply unit.

8.3. MEASUREMENT OUTPUT

AAD-CEMS-200 provides Modbus RTU over RS485 for remote communication.

8.4. SAMPLING PORT AT STACK

Sampling ports shall be provided by the user to fit the sample probe of the AAD-CEMS-200 to the stack.

8.5. SITE REQUIREMENT

AAD-CEMS-200 should be installed preferably in 24x7 air-conditioned environment or in fully shaded closed room free from dust with the temperature not exceeding 30 °C.

8.6. OTHER GENERAL REQUIREMENTS

8.6.1. ½” Instrument Air line of greater than 3 kg/cm2 pressure with isolation valve and air filter regulator is required for purging.

8.6.2. Warranty is void if failure due to “No air supply” and/or Surge/spike/interrupted power supply.

8.6.3. Instrument Air should be regulated and free from dust, moisture and oil mist.

8. MAINTENANCE

8.1. DOS AND DONTS

8.1.1. Always use Uninterrupted Power Supply (UPS). Do not connect the system to un-stabilized power supply. Instrument damaged due to fluctuations in power supply will not be covered under warranty.

8.1.2. Continuous compressed air supply for Purge air with pressures (4 bar to 5 bar) is necessary.

8.1.3. Do not switch ON the instrument without purge air supply.

8.1.4. Do keep the instrument free from dust, vibrations, heat, and electromagnetic noise generating sources.

8.1.5. Do keep one set of airline filters and regulators for purge air supply right before supplying to the instrument.

8.1.6. Do provide Weather shield cover for transmitter and receiver units.

8.1.7. Do keep the CEMS Analyzer in an air-conditioned room. Ensure the room temperature do not increase above 28 °C.

8.1.8. Do not try to open/repair/service the instrument in case of faulty measurements. Please contact customer support.

8.1.9. Do not look directly at the laser. It may cause serious eye injury, loss of vision, etc.

8.1.10. Do not touch the electrical terminals with bare hands.

8.1.11. The TFT/LCD screens may be damaged by sharp objects. Do not use any stylus, metal, wood objects to operate display screen.

8.2. PERIODIC MAINTENANCE

8.2.1. Daily Maintenance:

- 8.2.1.1. Check the compressed air supply for 4 to 5 bar air supply.
- 8.2.1.2. Check the measurement values for usual reading.
- 8.2.1.3. Check the rotameter (flowmeter) in the CEMS analyzer for 1 lpm sample gas flowrate. Do adjustments if needed.
- 8.2.1.4. Check the temperature controllers of Dehumidifier, Nox converter, heated hose are ON and are within the limits. (Refer Daily Checklist Document)

8.2.2. Monthly Maintenance:

- 8.2.2.1. In the stack, check the laser light from SPM Transmitter by inserting a white sheet of paper. The laser light dot must be within 6-7mm in diameter. Clean the transmitter and receiver optical components with extreme care.

8.2.2.2. Check the filter cartridge in the analyzer. If change of colour is observed, the filter cartridge shall be replaced with new one.

8.2.3. Quarterly Maintenance: Check the connection terminals for loose connection. If any disconnection/loose-connection observed, fasten the screws on the terminals to make a close-fitting connection.

8.2.4. Yearly Maintenance:

8.2.4.1. Check whether the range of the stack has changed.

8.2.4.2. Recalibrate the instrument.

8.2.4.3. Check for wear and tear in the connecting cables.

8.3. TROUBLESHOOTING

Some troubleshooting methods are listed below.

S. No	**Problem**	**Possible Causes**	**Troubleshooting**
1.	Instrument is not switching ON	1. Power supply failure 2. SMPS failure	1. Check the power supply to the system 2. Contact customer support
2.	Instrument switches ON. Readings are not updated.	1. Motherboard malfunction	1. Please contact customer support
3.	Temperature controllers not switched ON/not working	1. PLC issues	1. Please contact customer support
4.	Gas Measurement value –zero always	1. Calibration error	1. Perform Recalibration
5.	Gas Full-scale measurement always	1. Calibration error	1. Perform Recalibration
6.	SPM Measurement values higher than usual	1. Accumulation of dust on the transmitter and receiver optical components 2. Laser light out of focus 3. Transmitter/Receiver unit malfunction	1. Contact a trained person to clean the transmitter and receiver optical components. Check air supply. 2. Contact customer support 3. Contact customer support
7.	SPM Measurement values lower than usual	1. Dust concentration very low. 2. Lower particle size	1. Perform Manual reading & Recalibration